Natural History of Radiation Papillopathy after Proton Beam Irradiation of Parapapillary Melanoma

Ivana K. Kim, MD,[1] Anne Marie Lane, MPH,[1] Kathleen M. Egan, ScD,[2] John Munzenrider, MD,[3] Evangelos S. Gragoudas, MD[1]

Purpose: To evaluate the natural history of radiation papillopathy after proton beam irradiation of parapapillary choroidal melanoma.

Design: Noncomparative case series.

Participants: Ninety-three eyes of 93 patients.

Methods: Retrospective chart review of patients with choroidal melanoma within 0 to 1 disc diameter (DD) of the optic nerve and at least 2 DD away from the fovea treated with proton beam irradiation.

Main Outcome Measures: Rates of papillopathy, visual loss, and visual improvement.

Results: Sixty-eight percent of patients developed papillopathy a median of 1.5 years (17.7 months) after radiation. Among patients who developed papillopathy, 5-year rates of vision retention were 42% for counting fingers or better and 23% for 20/200 or better. At 5 years after the onset of papillopathy, 31% of patients had spontaneous visual improvement of 3 or more lines compared with the time of papillopathy diagnosis.

Conclusions: Even in eyes receiving high doses of radiation to the optic nerve for treatment of intraocular tumors, approximately two thirds of patients develop radiation papillopathy. After the development of papillopathy, 42% of eyes retain useful vision and one third of eyes have significant spontaneous visual improvement over 5 years. The natural history of radiation papillopathy may be more favorable than previously assumed, and therefore treatments for this condition must be compared with an appropriate control group.

Financial Disclosure(s): Proprietary or commercial disclosure may be found after the references.
Ophthalmology 2010;117:1617–1622 © 2010 by the American Academy of Ophthalmology.

Radiation papillopathy is an expected complication of irradiation of ocular tumors in the parapapillary area. Previous series have evaluated rates and risk factors for radiation papillopathy,[1–3] but the natural history of this optic neuropathy has not been well characterized. Several small series reporting the results of various treatments for this condition have been published,[4–7] but given the lack of control groups and clearly defined natural history, evaluation of efficacy has been difficult. In this analysis, we examined the course of visual changes caused by radiation papillopathy in a group of patients with parapapillary choroidal melanoma treated with proton beam irradiation.

Subjects and Methods

A waiver of informed consent and Health Insurance Portability and Accountability Act authorization was granted by the Institutional Review Board of the Massachusetts Eye and Ear Infirmary for this medical records review.

Patients with parapapillary tumors (N=93) treated with proton therapy at the Harvard Cyclotron Laboratory between 1985 and 1997 were identified through the Melanoma Registry of the Ocular Oncology Service at the Massachusetts Eye and Ear Infirmary, Boston, Massachusetts. Patients with choroidal melanomas located within 1 disc diameter (DD) of the optic nerve and at least 2 DD away from the fovea, measuring less than 8 mm in height and 15 mm in largest tumor diameter, were included in this analysis. Patients with these tumor characteristics were selected for evaluation because they were more likely to be at risk of vision loss caused by papillopathy than by other vision-compromising complications. In addition, analysis was restricted to patients who resided in the United States or Canada at the time of diagnosis and had completed at least 1 post-radiation ophthalmologic examination. All patients were followed for ocular outcomes, including radiation complications. Mean follow-up time was 5.5 years with a range of 6 months to 13 years. Mean follow-up for patients not developing radiation papillopathy was 4.4 years (range 6 months to 11.75 years). Evaluations were performed at 12 months after irradiation in 98% of patients, at 24 months after irradiation in 84% of patients, at 36 months after irradiation in 72% of patients, and at 60 months after irradiation in 53% of patients.

Patients were evaluated annually at minimum at the Massachusetts Eye and Ear Infirmary or by their local ophthalmologist. The ophthalmologic examination at each visit included Snellen visual acuity testing, slit-lamp biomicroscopy, indirect ophthalmoscopy, and fundus photography. A diagnosis of radiation papillopathy was made if there was evidence of nerve fiber layer infarcts, hemorrhage, exudate, edema, or pallor involving the disc. No patient received any treatment for radiation papillopathy.

The treatment planning protocol for proton therapy of uveal melanomas has been described.[8] A 3-dimensional treatment planning computer program facilitates the selection of the appropriate fixation angle, which is chosen to minimize irradiation of the lens, fovea, and optic disc. Patient immobilization is achieved via a bite

Published by Elsevier Inc.

ISSN 0161-6420/10/$–see front matter
doi:10.1016/j.ophtha.2009.12.015